Iframes/Popups Are Dangerous in Mobile WebView: Studying and Mitigating Differential Context Vulnerabilities

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Content

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Abstract:

In this paper, we present a novel class of Android WebView vulnerabilities (called Differential Context Vulnerabilities or DCVs) associated with web iframe/popup behaviors. To demonstrate the security implications of DCVs, we devise several novel concrete attacks. We show an untrusted web iframe/popup inside WebView becomes dangerous that it can launch these attacks to open holes on existing defense solutions, and obtain risky privileges and abilities, such as breaking web messaging integrity, stealthily accessing sensitive mobile functionalities, and performing phishing attacks.

Then, we study and assess the security impacts of DCVs on real-world apps. For this purpose, we develop a novel technique, DCV-Hunter, that can automatically vet Android apps against DCVs. By applying DCV-Hunter on a large number of most popular apps, we find DCVs are prevalent. Many high-profile apps are verified to be impacted, such as Facebook, Instagram, Facebook Messenger,

Google News, Skype, Uber, Yelp, and U.S. Bank. To mitigate DCVs, we design a multi-level solution that enhances the security of WebView. Our evaluation on real-world apps shows the mitigation solution is effective and scalable, with negligible overhead.

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